

TLM ports

UVM components talk through TLM ports and exports, not by reaching into each other's pins

Port, export, and the three patterns

- Think port connects to export
- Seq item is the classic pull path
- Put and get are push or pull styles between producer and consumer blocks
- Analysis is different, it is a broadcast
- You wire all of this in connect phase, after build has created the port objects

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

TLM port wiring

Sketch how components talk without pin wiggles: **seq_item port**→**export, put/get**, and **analysis port** → **subscriber**. Starter: driver connected to sequencer; monitor analysis wired to scoreboard.

Starter example: seq_item_port → sequencer export and monitor analysis_port → scoreboard are both connected.

Load starter example

Challenges 0 / 22 cleared

Quiz: connect: TLM ports are usually wired in...

☐ connect_phase (port.connect(export)) ☐ only after \$finish ☐ synthesis ☐ report_phase exclusively

Show hint

Check

Next

Idle

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Core ideas

port → export

Connect initiator port to a target export in connect_phase.

seq_item

Driver pulls items from the sequencer.

analysis

Monitor write() broadcasts to subscribers.

put / get

Push or pull a transaction across components.

Lab

SCENARIO

TLM KIND

TLM port wiring lab starter

Real UVM literacy

```
wsl - module08-uvm-tlm/examples/tlm-sketch

# Track A - real Unix
# real shell session (Track A - TLM port wiring)

$ pwd
/mnt/d/proj/designs/learning/courses/learn_uvm2017/module08-uvm-tlm

$ ls examples/tlm-sketch/
wiring.txt

$ cat examples/tlm-sketch/wiring.txt
UVM TLM wiring sketch (literacy)
=====

Rule: port (initiator) .connect( export (target) )
      Wire in connect_phase - after build creates port objects.

1) seq_item - driver pulls from sequencer
   drv.seq_item_port.connect( sqr.seq_item_export );
   // in driver run task:
   //   seq_item_port.get_next_item( txn );
   //   ... drive pins ...
   //   seq_item_port.item_done();

2) put - push from producer to consumer
   prod.put_port.connect( cons.put_export );
   // producer calls put_port.put( txn );

3) get - pull from producer export
   cons.get_port.connect( prod.get_export );
   // consumer calls get_port.get( txn );

4) analysis - monitor broadcast (one → many)
   mon.analysis_port.connect( sb.analysis_export ); // or uvm_analysis_imp
   // monitor calls analysis_port.write( txn );
   // subscribers receive write() - fan-out

Classic active agent hookup:
driver.seq_item_port.connect( sequencer.seq_item_export );
monitor.analysis_port.connect( scoreboard.actual_imp );

Common mistakes:
wiring in build_phase (ports may not exist yet)
reversed port/export direction
forget item_done after get_next_item
treating analysis like a paired export (it's broadcast)

$ grep -n "seq_item_port\|analysis_port\\\|\.connect("
.././learn_uvm2017_sv_verilator/module08/tests/uvm_tests/test_utilities_uvm.sv | head -10
80:         seq_item_port.get_next_item(txn);
83:         seq_item_port.item_done();
92:         uvm_analysis_port #(UtilitiesTransaction) analysis_port;
98:         analysis_port = new("analysis_port", this);
111:         analysis_port.write(txn);
198:         driver.seq_item_port.connect(sequencer.seq_item_export);
230:         agent.monitor.analysis_port.connect(scoreboard.actual_imp);
```

Real shell — TLM wiring sketch

Pitfalls to watch

- Do not wire ports in build, they may not exist yet; connect phase is the usual place
- Do not confuse port and export direction, initiator connects to target
- Do not forget item done after get next item or the sequencer stalls
- Analysis write is one-to-many, not a paired export like seq item
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, load starter, disconnect analysis, and explain what breaks
- On real UVM, sketch seq item and analysis links for one active agent
- When you are ready, take the short quiz, then continue to sequence flow in the next module